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Next item →

1. In the discrete-time state-space model:

1 / 1 point

$$x_{k+1} = Ax_k + Bu_k + w_k$$

$$z_k = Cx_k + Du_k + v_k,$$

which of the following is true?

- ☐ x_k is the system's state vector, u_k is a sensor measurement, w_k is the disturbance or process noise input, z_k is the deterministic input, and v_k is sensor noise.
- ☐ x_k is the deterministic input, u_k is the system's state vector, w_k is the disturbance or process noise input, z_k is a sensor measurement, and v_k is sensor noise.
- ☐ x_k is the system's state vector, u_k is the deterministic input, w_k is sensor noise, z_k is a sensor measurement, and v_k is the disturbance or process noise input.
- ☒ x_k is the system's state vector, u_k is the deterministic input, w_k is the disturbance or process noise input, z_k is a sensor measurement, and v_k is sensor noise.

✔ Correct
Yes, you are correct.

2. If our goal is to make an optimal estimate of x_k in the discrete-time state-space model:

1 / 1 point

$$x_{k+1} = Ax_k + Bu_k + w_k$$

$$z_k = Cx_k + Du_k + v_k,$$

why do we need feedback? (Select all that apply.)

- ☒ Because we rarely (if ever) know the initial state x_0 exactly and feedback helps to correct for an uncertain guess of x_0 over time.

✔ Correct
Yes. This is one reason that we need feedback.

- ☐ Because we can never measure the u_k signal and must use feedback to help correct for the uncertainty that is caused by this inability.

- ☒ Because the values of the disturbance signal w_k are not known, and feedback helps to correct for the uncertainty that this contributes to the x_k signal.

✔ Correct
Yes. This is one reason that we need feedback.

- ☒ Because we never know the values for the model's matrices A , B , C and/or D perfectly, and feedback helps to correct for errors in our model.

✔ Correct
Yes. This is one reason that we need feedback.

3. What is the operating principle of a discrete-time Kalman filter?

1 / 1 point

- ☐ Based on an initial state estimate $\hat{x}_0$, the algorithm repeatedly predicts process noise w_{k-1} and creates a state estimate by computing $\hat{x}_k = A\hat{x}_{k-1} + Bu_{k-1} + w_{k-1}$.
- ☐ Based on an initial state estimate $\hat{x}_0$, the algorithm repeatedly creates a state estimate by computing $\hat{x}_k = A\hat{x}_{k-1} + Bu_{k-1}$.
- ☐ Based on an initial state estimate $\hat{x}_0$, the algorithm then predicts the value of $\hat{x}_1^-$ and corrects that value to $\hat{x}_1^+$ after measuring z_1 . Then, the algorithm runs open-loop to make all other estimates $\hat{x}_k^+$.
- ☒ Based on an initial state estimate $\hat{x}_0$, the algorithm then predicts the value of $\hat{x}_1^-$ and corrects that value to $\hat{x}_1^+$ after measuring z_1 . Then, the algorithm repeatedly executes prediction and correction (estimation) steps for every measurement z_k .

✔ Correct
Yes, this is correct.